

Structured Entity-Relationship Model

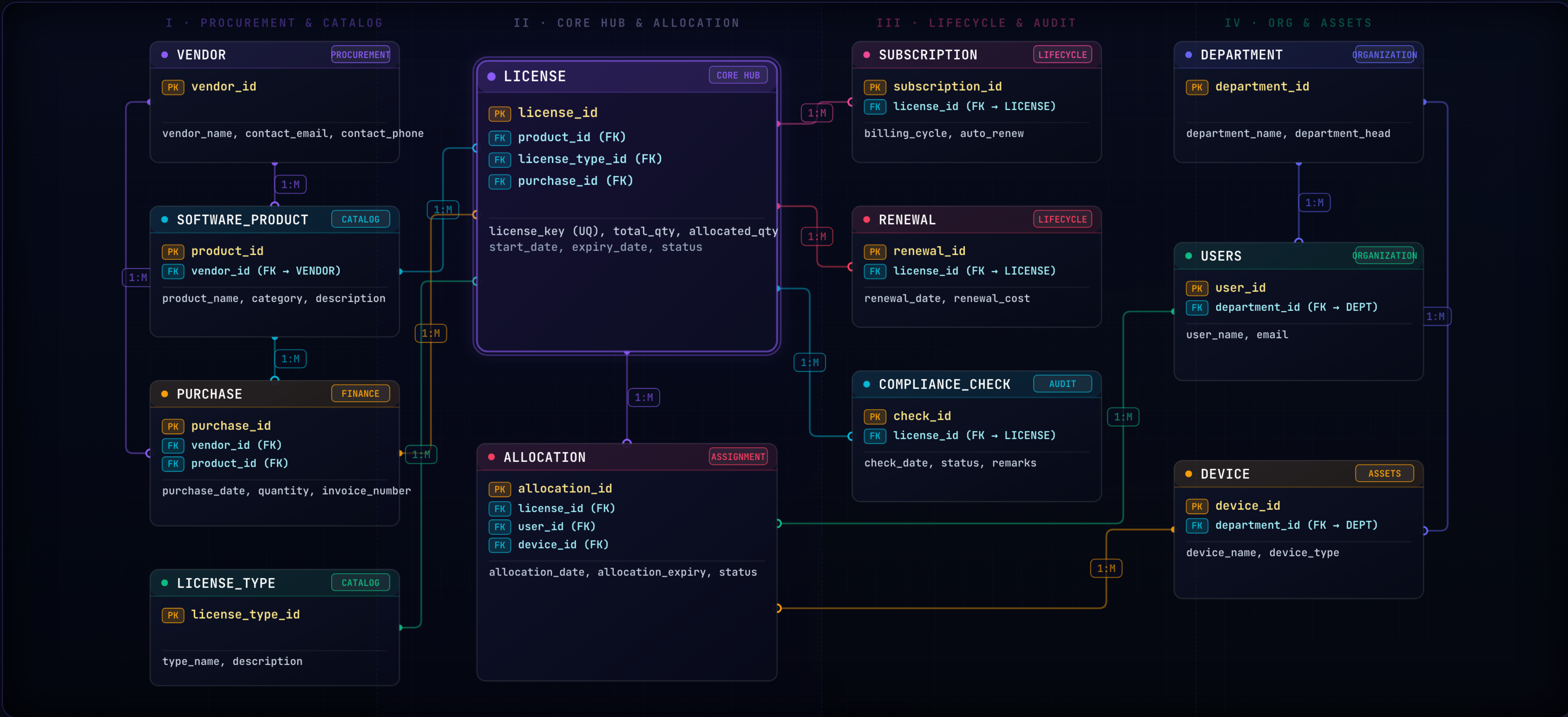
All Tables (12)

Procurement & Catalog (4)

Core Hub & Allocation (2)

Lifecycle & Audit (3)

Organization & Assets (3)



Architectural Legend · PK Primary Key · FK Foreign Key · 1:M One-to-Many Relational Cardinality

12 ENTITIES · 14 RELATIONSHIPS · 3NF

Cardinalities & Meanings

SOURCE	TARGET	CARDINALITY	MEANING
Vendor	Product	1 : M	One vendor supplies many software products
Product / License Type / Purchase	License	1 : M	Each entity generates multiple license records
License	Subscription	1 : M opt	Recurring billing for subscription-type licenses
License	Allocation	1 : M	One license (N seats) allocated to many users or devices
License	Renewal	1 : M	A license can be renewed multiple times over its lifetime
License	Compliance Check	1 : M	Compliance checks run repeatedly per license
Department	Users / Device	1 : M	A department owns many users and devices

Relational Schema, Part 1

• VENDOR		
vendor_id	INT	PK
vendor_name	VARCHAR(100)	NOT NULL
contact_email	VARCHAR(150)	UNIQUE
contact_phone	VARCHAR(20)	
address	TEXT	

• LICENSE_TYPE		
license_type_id	INT	PK
type_name	VARCHAR(50)	NOT NULL
description	TEXT	

• SOFTWARE_PRODUCT		
product_id	INT	PK
product_name	VARCHAR(150)	NOT NULL
vendor_id	INT	FK → VENDOR
category	VARCHAR(50)	
description	TEXT	

• PURCHASE		
purchase_id	INT	PK
vendor_id	INT	FK → VENDOR
product_id	INT	FK → PRODUCT
purchase_date	DATE	NOT NULL
quantity	INT	CHECK > 0
invoice_number	VARCHAR(50)	UNIQUE

PK Primary key FK Foreign key UNIQUE Unique constraint

Relational Schema, Part 2

● SUBSCRIPTION		
subscription_id	INT	PK
license_id	INT	FK → LICENSE
billing_cycle	VARCHAR(20)	
auto_renew	BOOLEAN	DEFAULT FALSE

● DEPARTMENT		
department_id	INT	PK
department_name	VARCHAR(100)	UNIQUE NOT NULL
department_head	VARCHAR(100)	

● USERS		
user_id	INT	PK
user_name	VARCHAR(100)	NOT NULL
email	VARCHAR(150)	UNIQUE NOT NULL
department_id	INT	FK → DEPT

● DEVICE		
device_id	INT	PK
device_name	VARCHAR(100)	NOT NULL
device_type	VARCHAR(50)	
department_id	INT	FK → DEPT

● ALLOCATION		
allocation_id	INT	PK
license_id	INT	FK → LICENSE
user_id	INT	FK nullable
allocation_expiry	DATE	CHECK ≥ alloc_date
status	VARCHAR(20)	DEFAULT 'Active'

● RENEWAL		
renewal_id	INT	PK
license_id	INT	FK → LICENSE
renewal_date	DATE	NOT NULL
renewal_cost	DECIMAL	CHECK > 0

● COMPLIANCE_CHECK		
check_id	INT	PK
license_id	INT	FK → LICENSE
check_date	DATE	NOT NULL
status	VARCHAR(20)	
remarks	TEXT	

The database enforces these rules

Not the application layer — the schema itself.

● Unique License Keys	UNIQUE ON license.license_key
● Allocation Within Seat Count	CHECK (allocated_qty ≤ total_qty)
● Valid Allocation Period	CHECK (expiry ≥ allocated_date)
● No Expired License Assignment	WHERE status = 'Active'
● Positive Renewal Cost	CHECK (renewal_cost > 0)
● Referential Integrity	FOREIGN KEY ... REFERENCES

↑ Click any rule to expand details

DDL Schema & Relational Constraints

Production MySQL 8.0 / PostgreSQL tables with referential integrity, check constraints, and performance indexes.

presentation2/ddl.sql

- Core Licenses & Products
- Allocations & Devices
- Departments & Users
- Renewals & Audit Logs

01_core_schema.sqlCopy SQL

```
-- CORE LICENSES & SOFTWARE PRODUCTS
CREATE TABLE licenses (
  license_id      INT AUTO_INCREMENT PRIMARY KEY,
  license_key     VARCHAR(128) NOT NULL UNIQUE,
  product_id     INT NOT NULL,
  license_type_id INT NOT NULL,
  purchase_id     INT,
  total_seats     INT NOT NULL DEFAULT 1,
  allocated_seats INT NOT NULL DEFAULT 0,
  purchase_date   DATE NOT NULL,
  expiry_date     DATE NOT NULL,
  unit_cost       DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  total_cost      DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  billing_cycle   ENUM('MONTHLY', 'ANNUALLY', 'ONE_TIME') DEFAULT 'ANNUALLY',
  auto_renew      BOOLEAN DEFAULT FALSE,
  status          ENUM('ACTIVE', 'EXPIRED', 'SUSPENDED', 'PENDING_RENEWAL') DEFAULT 'ACTIVE',
  -- Relational Integrity Constraints
  CONSTRAINT fk_license_product FOREIGN KEY (product_id)
    REFERENCES software_products(product_id) ON DELETE CASCADE,
  CONSTRAINT chk_allocated_seats CHECK (allocated_seats ≤ total_seats),
  CONSTRAINT chk_license_dates  CHECK (expiry_date ≥ purchase_date),
  CONSTRAINT chk_unit_cost      CHECK (unit_cost ≥ 0.00)
);

CREATE INDEX idx_licenses_prod_status ON licenses(product_id, status);
CREATE INDEX idx_licenses_expiry     ON licenses(expiry_date);
```

ENFORCED DATABASE INVARIANTS

Seat Invariant

CHECK (allocated_seats ≤ total_seats)

Database engine rejects any transaction that causes over-licensing.

Chronological Bounds

CHECK (expiry_date ≥ purchase_date)

Guarantees contract validity dates cannot be inverted.

Cascading Deletion

ON DELETE CASCADE

Removing a software product cleanly purges associated licenses without orphan rows.

⚡ Fully Executable Master Script Ready

Contains 12 normalized tables with MySQL & PostgreSQL compatibility.

Transactional Seat Lifecycle & Operations

ACID-compliant transactions ensuring seat counts never drift, corrupt, or deadlock during concurrent IT provisioning.

presentation2/dml.sql

- Seat Allocation Transaction
- Seat Reclamation / Offboarding
- Annual Contract Renewal
- Enterprise Seed Data Population

tx_allocate_seat.sqlCopy SQL

```
-- TRANSACTION 1: ATOMIC SEAT PROVISIONING
START TRANSACTION;

-- Step 1: Row-lock the license record to eliminate concurrent race conditions
SELECT license_id, total_seats, allocated_seats, status
FROM licenses
WHERE license_id = 1
FOR UPDATE;

-- Step 2: Capacity check validated (allocated_seats < total_seats)
-- Insert new allocation record mapping employee & workstation
INSERT INTO license_allocations (
  allocation_code, license_id, user_id, device_id,
  allocated_date, allocation_expiry, status, allocated_by
) VALUES (
  CONCAT('ALC-', DATE_FORMAT(NOW(), '%Y%m%d'), '-', LPAD(FLOOR(RAND() * 9999), 4, '0')),
  1, 2, 2,
  CURRENT_DATE(), DATE_ADD(CURRENT_DATE(), INTERVAL 1 YEAR), 'ACTIVE', 'IT Admin'
);

-- Step 3: Increment allocated seats atomically
UPDATE licenses
SET allocated_seats = allocated_seats + 1
WHERE license_id = 1;

COMMIT;
```

ACID TRANSACTION PROTOCOL

- 01

Row-Level Pessimistic Lock
SELECT FOR UPDATE prevents simultaneous transactions from oversubscribing the last remaining seat.
- 02

Junction Record Creation
Creates verified license_allocations mapping with immutable audit code and timestamps.
- 03

Atomic Counter Synchronization
Increments allocated_seats in parent table; rolls back on any violation.
- Zero Phantom Allocations Guaranteed**
Database-level row locks ensure concurrent IT admins cannot over-allocate seats.

Enterprise Reporting & Analytical Queries

Real-world SQL queries powering expiration alerts, capacity warnings, department chargebacks, and compliance audits.

presentation2/queries.sql

- 🔥 30-Day Expiration Alert
- 📉 ⚡ Seat Saturation (≥ 90%)
- 💰 🏢 Department Software Spend
- 🏢 📊 Vendor Spend Concentration
- 🛡️ 📋 Compliance & Audit Risks

analytical_query.sql Copy SQL

```
-- QUERY 1: 30-DAY LICENSE EXPIRATION ALERT & PIPELINE
SELECT
  l.license_id,
  v.vendor_name,
  sp.product_name,
  l.total_seats,
  l.allocated_seats,
  l.expiry_date,
  DATEDIFF(l.expiry_date, CURRENT_DATE()) AS days_left,
  l.total_cost AS renewal_cost,
  CASE
    WHEN DATEDIFF(l.expiry_date, CURRENT_DATE()) ≤ 7 THEN 'CRITICAL'
    WHEN DATEDIFF(l.expiry_date, CURRENT_DATE()) ≤ 15 THEN 'URGENT'
    ELSE 'UPCOMING'
  END AS urgency_level
FROM licenses l
JOIN software_products sp ON l.product_id = sp.product_id
JOIN vendors v ON sp.vendor_id = v.vendor_id
WHERE l.status IN ('ACTIVE', 'PENDING_RENEWAL')
  AND l.expiry_date BETWEEN CURRENT_DATE() AND DATE_ADD(CURRENT_DATE(), INTERVAL 30 DAY)
ORDER BY l.expiry_date ASC;
```

RENEWAL PIPELINE
\$46,525 at risk [Live SQL View](#)

SIMULATED QUERY EXECUTION RESULT

PRODUCT	VENDOR	SEATS	EXPIRY DATE	DAYS LEFT	RENEWAL COST	URGENCY
Adobe Creative Cloud	Adobe Inc.	24 / 25	2026-02-01	14 days	\$21,575	URGENT
IntelliJ IDEA Ultimate	JetBrains	48 / 50	2026-02-15	28 days	\$24,950	UPCOMING

Tools, Stack & Timeline

● TECHNOLOGY STACK

● DATABASE

MySQL / PostgreSQL

● BACKEND

Java (JDBC + Swing) or Python (Tkinter/Flask)

● DIAGRAMS

draw.io — ER diagram export

● VERSION CONTROL

GitHub — feature-branch workflow

● DELIVERY TIMELINE		
✓ Review 1	Week 7	Problem definition, ER diagram, initial schema
✓ Review 2	Week 11	3NF schema, DDL, sample data, SQL queries & views (Delivered)
○ Review 3	Week 16	Working application, final demo, report, viva

Thank you.

QUESTIONS WELCOME